

7.29 From the definition of the angular momentum, $\vec{L} = \frac{1}{2\mu c^2} \int [\vec{r} \times (\vec{E} \times \vec{B}^*)] d^3x$, we can show

that

$$\frac{\vec{E} \times \vec{B}^*}{\sqrt{\mu\epsilon}} = \begin{vmatrix} \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \\ E_0 & \pm i E_0 & \frac{i}{k} \left(\frac{\partial E_0}{\partial x} \pm i \frac{\partial E_0}{\partial y} \right) \\ \pm i E_0 & E_0 & \pm \frac{1}{k} \left(\frac{\partial E_0}{\partial x} \mp i \frac{\partial E_0}{\partial y} \right) \end{vmatrix}$$

$$= \vec{e}_1 \left(i \frac{E_0}{k} \left(\frac{\partial E_0}{\partial x} \mp i \frac{\partial E_0}{\partial y} \right) - i \frac{E_0}{k} \left(\frac{\partial E_0}{\partial x} \pm i \frac{\partial E_0}{\partial y} \right) \right)$$

$$+ \vec{e}_2 \left(\mp \frac{E_0}{k} \left(\frac{\partial E_0}{\partial x} \pm i \frac{\partial E_0}{\partial y} \right) \mp \frac{E_0}{k} \left(\frac{\partial E_0}{\partial x} \mp i \frac{\partial E_0}{\partial y} \right) \right) + \vec{e}_3 \cdot 2E_0^2$$

$$= \vec{e}_1 \left(\pm \frac{2E_0}{k} \frac{\partial E_0}{\partial y} \right) + \vec{e}_2 \left(\mp \frac{2E_0}{k} \frac{\partial E_0}{\partial x} \right) + \vec{e}_3 \cdot 2E_0^2$$

and

$$\frac{\vec{r} \times (\vec{E} \times \vec{B}^*)}{\sqrt{\mu\epsilon}} = \begin{vmatrix} \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \\ x & y & z \\ \pm \frac{2E_0}{k} \frac{\partial E_0}{\partial y} & \mp \frac{2E_0}{k} \frac{\partial E_0}{\partial x} & 2E_0^2 \end{vmatrix}$$

$$= \vec{e}_1 \left(zy E_0^2 \pm 2z \frac{E_0}{k} \frac{\partial E_0}{\partial x} \right) + \vec{e}_2 \left(\pm 2z \frac{E_0}{k} \frac{\partial E_0}{\partial y} - 2xE_0^2 \right)$$

$$+ \vec{e}_3 \left(\mp \frac{2E_0}{k} \left(x \frac{\partial E_0}{\partial x} + y \frac{\partial E_0}{\partial y} \right) \right)$$

Therefore,

$$L_3 = \frac{\sqrt{\mu\epsilon}}{2\mu c^2} \int \left[\mp \frac{2E_0}{k} \left(x \frac{\partial E_0}{\partial x} + y \frac{\partial E_0}{\partial y} \right) \right] d^3x$$

$$= \mp \int \frac{\epsilon}{\mu} \frac{1}{2kc^2} \int \left(x \frac{\partial E_0^2}{\partial x} + y \frac{\partial E_0^2}{\partial y} \right) d^3x$$

$$= \mp \int \frac{\epsilon}{\mu} \frac{1}{kc^2} \int E_0(x, y)^2 d^3x,$$

where we have used integration by parts in the last step and dropped surface term.

Also,

$$U = \frac{1}{4} \int \left(\epsilon \vec{E} \cdot \vec{E}^* + \frac{1}{\mu} \vec{B} \cdot \vec{B}^* \right) d^3x = \frac{1}{4} \int (\epsilon \cdot 2E_0^2 + \epsilon \cdot 2E_0^2) d^3x = \epsilon \int E_0(x, y)^2 d^3x.$$

Since $\frac{1}{c} = \sqrt{\mu\epsilon}$, we have

$$\frac{L_3}{U} = \mp \frac{1}{kc} = \mp \frac{1}{\omega}.$$

Quantum mechanically, photons are spin-one massless particles, and their spin projections in the travelling direction are $\pm \hbar$. Photon energy is given by $\hbar\omega$. Thus, $L_3/\hbar = \pm \frac{1}{\omega}$.

Now consider the $\vec{L}_1$ -component.

$$\vec{L}_1 = \frac{\sqrt{\mu\epsilon}}{2\mu c^2} \int \left[2yE_0^2 \pm 2z \frac{E_0}{k} \frac{\partial E_0}{\partial x} \right] d^3x$$

If the wave is cylindrically symmetric, the first term is zero by symmetry. By integration

by parts, the second term is

$$\iint \left[\int \frac{z}{k} d(E_0^2) \right] dy dz = \iint \left[\frac{z}{k} E_0^2 \Big|_{x_{\min}}^{x_{\max}} - \int E_0^2 \frac{dz}{dx} dx \right] dy dz$$

If the wave is finite, E_0 will drop to zero at $x_{\min}$ and $x_{\max}$. The last term is 0, as

$\frac{dz}{dx} = 0$. Therefore, there will be no angular momentum component in the transverse direction for cylindrically symmetric, finite plane wave.